

GROUP 1

Loan Approval Prediction

Track	Machine Learning
Duration	2 Weeks (10 working days, 4 hours/day)
ML Task	Classification
Classification Type	Binary Classification
Target	Loan_Status: Y = Approved, N = Not Approved
User Interface	Required

Project Analysis

Financial institutions receive many loan applications. Manual evaluation can be slow and inconsistent. This project requires students to build a machine-learning application that analyzes applicant information and predicts whether a loan application is likely to be approved.

Students must understand the dataset, prepare the data, compare multiple classification models, select the best model, and connect the final model to a functional user interface.

What the Student Must Build (Mandatory)

- Import and inspect the loan dataset.
- Handle missing values and duplicates.
- Encode categorical features correctly.
- Perform Exploratory Data Analysis (EDA).
- Separate features from the target variable Loan_Status.
- Split data into training and testing sets.
- Train at least two classification models.
- Evaluate and compare the models using suitable classification metrics.
- Select and save the best model.
- Allow a user to enter new applicant information.
- Predict Approved / Not Approved.
- Display a confidence/probability score if supported.
- Include a functional user interface.

Technical Requirements

- Python
- Pandas
- NumPy
- Scikit-learn
- Joblib/Pickle
- Git & GitHub

- Streamlit, Gradio, Flask, or another suitable UI framework

Dataset Requirements

- Use the instructor-provided Loan Approval dataset.
- Target variable: Loan_Status.
- Classes: Y = Approved; N = Not Approved.
- Main inputs may include Gender, Married, Dependents, Education, Self_Employed, ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term, Credit_History, and Property_Area.

Optional Features (Bonus)

These features may improve the project but are not required for basic project completion.

- Feature importance visualization
- Loan approval analytics dashboard
- PDF prediction report
- Batch prediction using CSV

References and Starting Resources

- Kaggle - Loan Prediction Dataset
- Scikit-learn Documentation
- Pandas Documentation